

PROGRAM 1

```

GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);

    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if (pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }

    return 0;
}

```

```

sri_dhruthi@Dhruthi:~$ nano fork_command.c
sri_dhruthi@Dhruthi:~$ gcc fork_command.c -o fork_command
sri_dhruthi@Dhruthi:~$ ./fork_command
Enter Linux command: ls
Parent PID : 479
Child PID : 482
OSLab      exec_example  filename.c  fork_command  fork_exec.c  parent_child  pid.c  wait_example
display_pid exec_example.c  fork        fork_command.c  hello        parent_child.c  username  wait_example.
display_pid.c filename        fork.c      fork_exec      hello.c      pid            username.c
Child process completed.

```

Try uname, lscpu, lsblk, ps, top commands after this program

```
sri_dhruthi@Dhruthi: ~  
sri_dhruthi@Dhruthi:~$ ./fork_command  
Enter Linux command: uname  
Parent PID : 583  
Child PID : 584  
Linux  
Child process completed.  
sri_dhruthi@Dhruthi:~$ ./fork_command  
Enter Linux command: lscpu  
Parent PID : 596  
Child PID : 597  
Architecture:          x86_64  
CPU op-mode(s):        32-bit, 64-bit  
Address sizes:         39 bits physical, 48 bits virtual  
Byte Order:            Little Endian  
CPU(s):                12  
On-line CPU(s) list:   0-11  
Vendor ID:             GenuineIntel  
Model name:            13th Gen Intel(R) Core(TM) i5-1335U  
CPU family:            6  
Model:                 186  
Thread(s) per core:    2  
Core(s) per socket:    6  
Socket(s):             1  
Stepping:              3  
BogoMIPS:              4991.99  
Flags:                 fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush  
cp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_kn  
sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand h  
bpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bml avx2  
opt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnmi umip waitpkg gfni v  
clear serialize ibt flush_lld arch_capabilities  
  
Virtualization features:  
Virtualization:        VT-x  
Hypervisor vendor:     Microsoft  
Virtualization type:   full  
Caches (sum of all):  
L1d:                   288 KiB (6 instances)  
L1i:                   192 KiB (6 instances)  
L2:                    7.5 MiB (6 instances)  
L3:                    12 MiB (1 instance)  
NUMA:
```

```
sri_dhruthi@Dhruthi: ~  
sri_dhruthi@Dhruthi:~$ ./fork_command  
Enter Linux command: lsblk  
Parent PID : 611  
Child PID : 612  
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS  
sda   8:0      0 356.9M 1 disk  
sdb   8:16     0 159.4M 1 disk  
sdc   8:32     0    2G   0 disk [SWAP]  
sdd   8:48     0    1T   0 disk /mnt/wslg/distro  
/  
Child process completed.  
sri_dhruthi@Dhruthi:~$ ./fork_command  
Enter Linux command: ps  
Parent PID : 624  
Child PID : 625  
  PID TTY          TIME CMD  
  319 pts/0        00:00:00 bash  
  624 pts/0        00:00:00 fork_command  
  625 pts/0        00:00:00 ps  
Child process completed.  
sri_dhruthi@Dhruthi:~$ ./fork_command  
Enter Linux command: top  
Parent PID : 635  
Child PID : 638  
top - 09:33:20 up 11 min, 1 user, load average: 0.02, 0.02, 0.00  
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st  
MiB Mem : 7762.6 total, 7194.1 free, 556.1 used, 163.1 buff/cache  
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7206.5 avail Mem  


| PID | USER     | PR | NI | VIRT  | RES   | SHR   | S | %CPU | %MEM | TIME+   | COMMAND         |
|-----|----------|----|----|-------|-------|-------|---|------|------|---------|-----------------|
| 638 | sri_dhr+ | 20 | 0  | 8172  | 6020  | 3900  | R | 0.3  | 0.1  | 0:00.11 | top             |
| 1   | root     | 20 | 0  | 24180 | 15324 | 11544 | S | 0.0  | 0.2  | 0:00.86 | systemd         |
| 2   | root     | 20 | 0  | 3180  | 2208  | 2072  | S | 0.0  | 0.0  | 0:00.01 | init-systemd(Ub |
| 7   | root     | 20 | 0  | 3180  | 2044  | 1940  | S | 0.0  | 0.0  | 0:00.00 | init            |
| 50  | root     | 19 | -1 | 50424 | 16756 | 15472 | S | 0.0  | 0.2  | 0:00.20 | systemd-journal |
| 82  | systemd+ | 20 | 0  | 22540 | 14580 | 11956 | S | 0.0  | 0.2  | 0:00.09 | systemd-resolve |
| 89  | root     | 20 | 0  | 35372 | 12344 | 9248  | S | 0.0  | 0.2  | 0:00.45 | systemd-udevd   |
| 108 | root     | 20 | 0  | 2888  | 1948  | 1820  | S | 0.0  | 0.0  | 0:00.04 | chronyd-starter |
| 109 | root     | 20 | 0  | 4472  | 3172  | 2920  | S | 0.0  | 0.0  | 0:00.00 | cron            |
| 110 | message+ | 20 | 0  | 8816  | 5356  | 4676  | S | 0.0  | 0.1  | 0:00.04 | dbus-daemon     |


```

PROGRAM 2

```
GNU nano 8.7.1
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if (src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while ((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```

```
sri_dhruthi@Dhruthi:~$ nano file_copy.c
sri_dhruthi@Dhruthi:~$ gcc file_copy.c -o file_copy
sri_dhruthi@Dhruthi:~$ nano input.txt
sri_dhruthi@Dhruthi:~$ ./file_copy
File copied successfully.
sri_dhruthi@Dhruthi:~$ cat output.txt
Operating System Lab
File Copy using System Calls

sri_dhruthi@Dhruthi:~$ |
```